

SQL PROJECT- RETAIL SALES ANALYSIS

1. Open SQL Server Management Studio (SSMS) app, connect to the localhost or server, then create a database

```
--Creating a database named Retail_Sales CREATE DATABASE  
[Retail_Sales];
```

2. Import the data

Steps:

Right click on the database "Retail_Sales" → Task → Import flat file (the file was downloaded as a csv file), this brings the Import wizard interface → click Next → input the file name "SQL - Retail Sales Analysis - SQL - Retail Sales Analysis.csv" → Next (previews the first 50 rows) → Next (brings 'modify columns' interface; selected transactions_id as the primary key with the data type as int, sale_date as date data type, sale_time as time(0), customer_id as tinyint, gender as nvarchar(10), age as tinyint, category as nvarchar(50), quantiy as tinyint, price_per_unit as smallint, cogs decimal(10,0), total_sale as int) → Next → Finish → (inserted the data successfully) Close.

The columns datatypes

	Column Name	Data Type	Allow Nulls
🔑	transactions_id	int	☐
	sale_date	date	☐
	sale_time	time(0)	☐
	customer_id	tinyint	☐
	gender	nvarchar(10)	☐
	age	tinyint	☒
	category	nvarchar(50)	☐
	quantiy	tinyint	☒
	price_per_unit	smallint	☒
	cogs	decimal(10, 0)	☒
	total_sale	int	☒

3. Perform Exploratory Data Analysis (using syntax)

A. Explore all object and columns in the database

```
-- Explore all object in the database
```

```
SELECT * FROM INFORMATION_SCHEMA.TABLES
```

```
-- Explore all columns in the Database
```

```
SELECT * FROM INFORMATION_SCHEMA.COLUMNS
```

```
WHERE TABLE_NAME = 'SQL - Retail Sales Analysis - SQL - Retail Sales Analysis'
```

B. Count total rows/records

```
-- B. Count rows
```

```
SELECT COUNT(*) AS TotalRows FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis] ;
```

The query ran successfully and gave TotalRows = 2000

C. To view the data

-- C. To view the data

```
SELECT TOP (1000) [transactions_id]
, [sale_date]
, [sale_time]
, [customer_id]
, [gender]
, [age]
, [category]
, [quantiy]
, [price_per_unit]
, [cogs]
, [total_sale]
FROM [Retail_Sales].[dbo].[SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
```

4. Data Cleaning

i. Identify and remove any records with missing or null values.

-- Identify records with missing or null values.

```
SELECT
SUM(CASE WHEN transactions_id IS NULL THEN 1 ELSE 0 END) AS
Null_transaction_id,
SUM(CASE WHEN sale_date IS NULL THEN 1 ELSE 0 END) AS Null_sales_date,
SUM(CASE WHEN sale_time IS NULL THEN 1 ELSE 0 END) AS Null_sales_time,
SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END) AS Null_customer_id,
SUM(CASE WHEN gender IS NULL THEN 1 ELSE 0 END) AS Null_gender,
SUM(CASE WHEN age IS NULL THEN 1 ELSE 0 END) AS Null_age,
SUM(CASE WHEN category IS NULL THEN 1 ELSE 0 END) AS Null_category,
SUM(CASE WHEN quantiy IS NULL THEN 1 ELSE 0 END) AS Null_quantiy,
SUM(CASE WHEN price_per_unit IS NULL THEN 1 ELSE 0 END) AS
Null_price_per_unit,
SUM(CASE WHEN cogs IS NULL THEN 1 ELSE 0 END) AS Null_cogs,
SUM(CASE WHEN total_sale IS NULL THEN 1 ELSE 0 END) AS Null_total_sale FROM
[SQL - Retail Sales Analysis - SQL - Retail Sales Analysis];
```

Only age has 10 null records and quantiy, price_per_unit, cogs, total_sale all has 3 null records.

ii. Replace and delete null values/records

-- Since the age with null records has useful information on transaction details it's better to replace the Null with 'None' by flagging it with an additional column.

```
ALTER TABLE dbo.[SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
ADD age_flag VARCHAR(10);

UPDATE dbo.[SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
SET age_flag = CASE
WHEN NULLIF(TRIM(CAST(age AS VARCHAR(10))), '') IS NULL THEN 'None'
ELSE 'Known'
END;
```

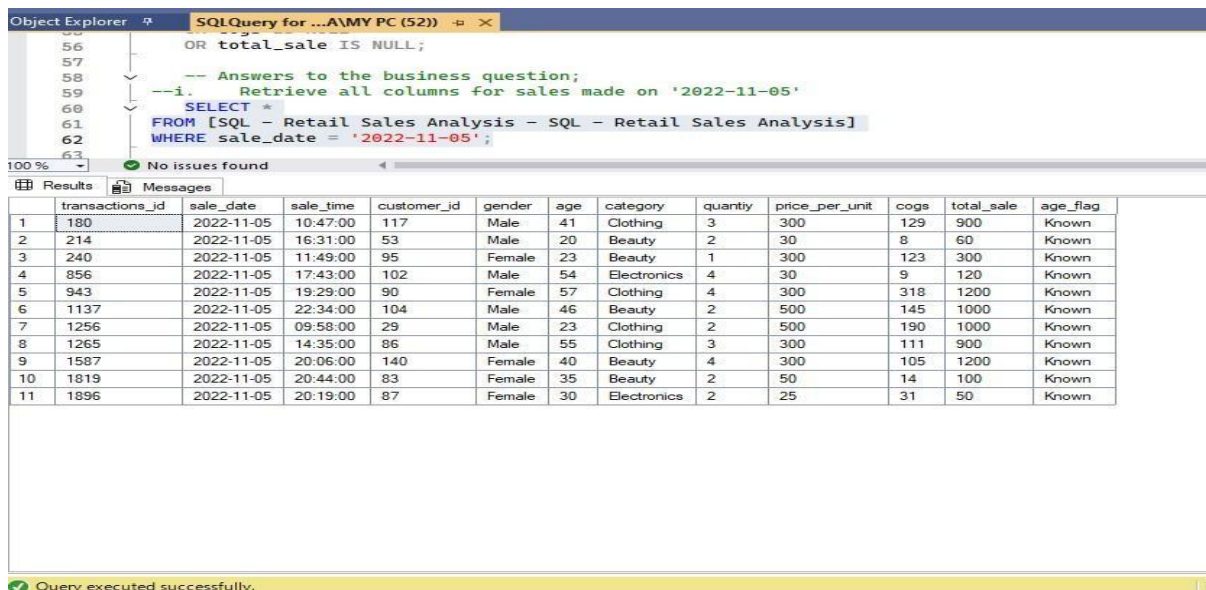
iii. Delete null records for quantiy, price_per_unit, cogs and total_sale

```
DELETE FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
WHERE quantiy IS NULL
    OR price_per_unit IS NULL
    OR cogs IS NULL
    OR total_sale IS NULL;
```

Answers to the business question;

i. Retrieve all columns for sales made on '2022-11-05'

```
SELECT *
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
WHERE sale_date = '2022-11-05';
```



The screenshot shows a SQL Server Enterprise Manager interface. The top pane displays a query window titled 'SQLQuery for ...\MY PC (52))'. The query is as follows:

```
56 OR total_sale IS NULL;
57
58 -- Answers to the business question;
59 --i. Retrieve all columns for sales made on '2022-11-05'
60 SELECT *
61 FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
62 WHERE sale_date = '2022-11-05';
63
```

The bottom pane shows the results of the query, displaying a table with 11 rows and 12 columns. The columns are: transactions_id, sale_date, sale_time, customer_id, gender, age, category, quantiy, price_per_unit, cogs, total_sale, and age_flag. The data is as follows:

	transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantiy	price_per_unit	cogs	total_sale	age_flag
1	180	2022-11-05	10:47:00	117	Male	41	Clothing	3	300	129	900	Known
2	214	2022-11-05	16:31:00	53	Male	20	Beauty	2	30	8	60	Known
3	240	2022-11-05	11:49:00	95	Female	23	Beauty	1	300	123	300	Known
4	856	2022-11-05	17:43:00	102	Male	54	Electronics	4	30	9	120	Known
5	943	2022-11-05	19:29:00	90	Female	57	Clothing	4	300	318	1200	Known
6	1137	2022-11-05	22:34:00	104	Male	46	Beauty	2	500	145	1000	Known
7	1256	2022-11-05	09:58:00	29	Male	23	Clothing	2	500	190	1000	Known
8	1265	2022-11-05	14:35:00	86	Male	55	Clothing	3	300	111	900	Known
9	1587	2022-11-05	20:06:00	140	Female	40	Beauty	4	300	105	1200	Known
10	1819	2022-11-05	20:44:00	83	Female	35	Beauty	2	50	14	100	Known
11	1896	2022-11-05	20:19:00	87	Female	30	Electronics	2	25	31	50	Known

The status bar at the bottom indicates 'Query executed successfully.'

ii. Transactions where category is 'Clothing' and quantity > 4 in Nov-2022

```
SELECT *
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
WHERE category = 'Clothing'
    AND quantiy > 4
    AND sale_date >= '2022-11-01'
    AND sale_date < '2022-12-01';
```

Object Explorer: SQLQuery for ...\AMY PC (52))

```

-- ii. Retrieve all transactions where category is 'Clothing' and quantity > 4 in Nov-2022
SELECT *
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
WHERE category = 'Clothing'
AND quantity > 4
AND sale_date >= '2022-11-01'
AND sale_date < '2022-12-01';

```

Results: No issues found

transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale	age_flag
-----------------	-----------	-----------	-------------	--------	-----	----------	----------	----------------	------	------------	----------

Query executed successfully.

There was no transaction where category is 'Clothing' and quantity > 4 in Nov-2022

iii. Calculate total sales for each category

```

SELECT category, SUM(total_sale) AS total_sales
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
GROUP BY category
ORDER BY SUM(total_sale) DESC; -- use the aggregate function, not the alias

```

Object Explorer: SQLQuery for ...\AMY PC (52))

```

-- iii. Calculate total sales for each category
SELECT category, SUM(total_sale) AS total_sales
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
GROUP BY category
ORDER BY SUM(total_sale) DESC; -- use the aggregate function, not the alias

```

Results: No issues found

category	total_sales
1 Electronics	313810
2 Clothing	311070
3 Beauty	286840

Query executed successfully.

iv. Find the average age of customers who purchased items from 'Beauty' (using age_flag)

```

SELECT ROUND(AVG(age), 2) AS Avgage_Beauty
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
WHERE category = 'Beauty'
AND age_flag = 'Known'; ---- only includes rows with valid age

```

Object Explorer SQLQuery for ...\MY PC (52) x

```

78 -- iv. Find the average age of customers who purchased items from 'Beauty' (using age_flag)
79 SELECT ROUND(AVG(age), 2) AS Avgage_Beauty
80 FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
81 WHERE category = 'Beauty'
82 AND age_flag = 'Known'; ----- only includes rows with valid age
83

```

100 % No issues found

Results Messages

	Avgage_Beauty
1	40

Query executed successfully.

v. Find all transactions where the total_sale is greater than 1000.

```

SELECT *
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis] WHERE total_sale
> 1000;

```

Object Explorer SQLQuery for ...\MY PC (52) x

```

83
84 -- v. Find all transactions where total_sale > 1000 Find all transactions where TotalSales > 1000
85 SELECT *
86 FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
87 WHERE total_sale > 1000;
88

```

100 % No issues found

Results Messages

	transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale	age_flag
1	13	2023-02-08	17:43:00	106	Male	22	Electronics	3	500	245	1500	Known
2	15	2022-07-01	11:50:00	75	Female	42	Electronics	4	500	210	2000	Known
3	16	2022-06-25	10:33:00	82	Male	19	Clothing	3	500	180	1500	Known
4	31	2023-12-31	17:47:00	3	Male	44	Electronics	4	300	129	1200	Known
5	46	2022-11-08	17:50:00	54	Female	20	Electronics	4	300	84	1200	Known
6	47	2022-10-22	17:22:00	96	Female	40	Beauty	3	500	600	1500	Known
7	54	2022-10-20	10:17:00	142	Female	38	Electronics	3	500	200	1500	Known
8	58	2023-09-16	19:18:00	53	Male	18	Clothing	4	300	75	1200	Known
9	65	2022-12-11	20:03:00	84	Male	51	Electronics	4	500	160	2000	Known
10	67	2023-08-19	20:19:00	119	Female	48	Beauty	4	300	129	1200	Known
11	72	2023-12-06	19:19:00	5	Female	20	Electronics	4	500	195	2000	Known
12	74	2023-10-05	19:50:00	56	Female	18	Beauty	4	500	205	2000	Known
13	78	2023-02-17	21:08:00	68	Female	47	Clothing	3	500	265	1500	Known
14	89	2023-12-30	21:15:00	117	Female	55	Electronics	4	500	590	2000	Known
15	93	2022-01-25	20:52:00	148	Female	35	Beauty	4	500	140	2000	Known
16	99	2023-11-19	15:12:00	71	Female	50	Electronics	4	300	132	1200	Known
17	107	2022-10-06	09:18:00	75	Female	21	Clothing	4	300	78	1200	Known
18	109	2023-09-06	19:57:00	94	Female	34	Electronics	4	500	560	2000	Known
19	111	2023-04-15	09:45:00	5	Female	34	Electronics	3	500	130	1500	Known
20	112	2023-12-25	18:44:00	57	Male	37	Clothing	3	500	165	1500	Known
21	115	2022-09-02	19:21:00	67	Male	51	Clothing	3	500	255	1500	Known
22	118	2023-03-13	20:07:00	3	Female	30	Electronics	4	500	270	2000	Known

Query executed successfully.

vi. Find the total number of transactions(transaction_id) made by each gender in each category

```

SELECT gender, category, COUNT(transactions_id) AS total_transactions
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
GROUP BY gender, category
ORDER BY category, gender;

```

Object Explorer SQLQuery for ...\AMY PC (52)

```

89 -- vi.Find the total number of transactions(transaction_id) made by each gender in each category
90 SELECT gender, category, COUNT(transaction_id) AS total_transactions
91 FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
92 GROUP BY gender, category
93 ORDER BY category, gender;
94

```

100 % No issues found

	gender	category	total_transactions
1	Female	Beauty	330
2	Male	Beauty	282
3	Female	Clothing	347
4	Male	Clothing	354
5	Female	Electronics	340
6	Male	Electronics	344

Query executed successfully.

Vii. Calculate the average sale for each month and find out the best-selling month in each year

-- Monthly average sales

```

SELECT
    YEAR(sale_date) AS sale_year,
    MONTH(sale_date) AS sale_month,
    AVG(total_sale) AS Avgmonthly_sales
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
GROUP BY YEAR(sale_date), MONTH(sale_date)
ORDER BY sale_year, sale_month;

```

Object Explorer SQLQuery for ...\AMY PC (52)

```

94 -- vii.Calculate the average sale for each month and find out the best-selling month in each year
95 -- Monthly average sales
96 SELECT
97     YEAR(sale_date) AS sale_year,
98     MONTH(sale_date) AS sale_month,
99     AVG(total_sale) AS Avgmonthly_sales
100 FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
101 GROUP BY YEAR(sale_date), MONTH(sale_date)
102 ORDER BY sale_year, sale_month;
103

```

100 % No issues found

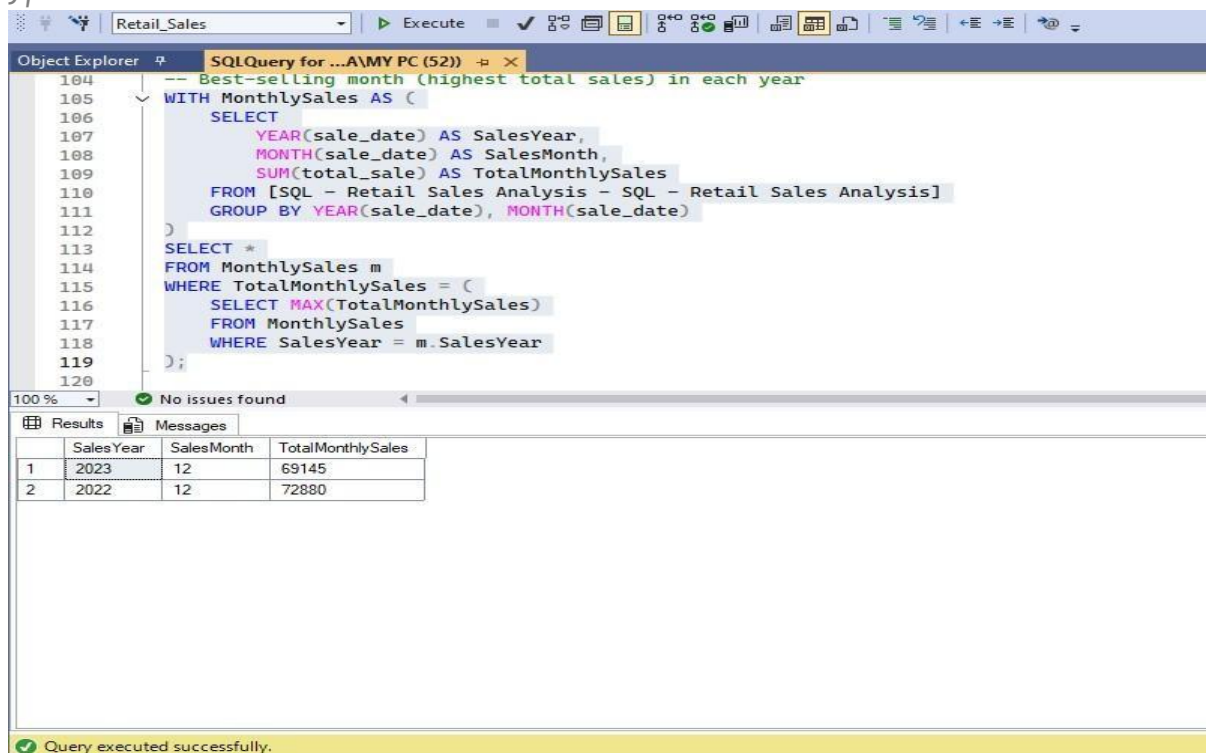
	sale_year	sale_month	Avgmonthly_sales
1	2022	1	397
2	2022	2	366
3	2022	3	521
4	2022	4	486
5	2022	5	480
6	2022	6	481
7	2022	7	541
8	2022	8	385
9	2022	9	478
10	2022	10	467
11	2022	11	472
12	2022	12	464
13	2023	1	396
14	2023	2	535
15	2023	3	394
16	2023	4	466
17	2023	5	450
18	2023	6	438
19	2023	7	427
20	2023	8	495

Query executed successfully.

```

-- Best-selling month (highest total sales) in each year
WITH MonthlySales AS (
    SELECT
        YEAR(sale_date) AS SalesYear,
        MONTH(sale_date) AS SalesMonth,
        SUM(total_sale) AS TotalMonthlySales
    FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
    GROUP BY YEAR(sale_date), MONTH(sale_date)
)
SELECT *
FROM MonthlySales m
WHERE TotalMonthlySales = (
    SELECT MAX(TotalMonthlySales)
    FROM MonthlySales
    WHERE SalesYear = m.SalesYear
);

```



Object Explorer: SQLQuery for ...A\MY PC (52)

```

104 -- Best-selling month (highest total sales) in each year
105 WITH MonthlySales AS (
106     SELECT
107         YEAR(sale_date) AS SalesYear,
108         MONTH(sale_date) AS SalesMonth,
109         SUM(total_sale) AS TotalMonthlySales
110     FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
111     GROUP BY YEAR(sale_date), MONTH(sale_date)
112 )
113 SELECT *
114 FROM MonthlySales m
115 WHERE TotalMonthlySales = (
116     SELECT MAX(TotalMonthlySales)
117     FROM MonthlySales
118     WHERE SalesYear = m.SalesYear
119 );
120

```

100 % No issues found

	SalesYear	SalesMonth	TotalMonthlySales
1	2023	12	69145
2	2022	12	72880

Query executed successfully.

viii. Find the top 5 customers based on highest total sales

```

SELECT TOP 5 customer_id, SUM(total_sale) AS total_spent
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
GROUP BY customer_id
ORDER BY total_spent DESC;

```

Object Explorer SQLQuery for ...A\MY PC (52)

```

120
121 -- viii. Find the top 5 customers based on highest total sales
122 SELECT TOP 5 customer_id, SUM(total_sale) AS total_spent
123 FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
124 GROUP BY customer_id
125 ORDER BY total_spent DESC;

```

100 % No issues found

Results Messages

	customer_id	total_spent
1	3	38440
2	1	30750
3	5	30405
4	2	25295
5	4	23580

Query executed successfully.

ix. Find the number of unique customers who purchased items from each category

```

SELECT category, COUNT(DISTINCT customer_id) AS UniqueCustomers
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
GROUP BY category
ORDER BY UniqueCustomers DESC;

```

Object Explorer SQLQuery for ...A\MY PC (52)

```

126
127 -- ix Find the number of unique customers who purchased items from each category
128 SELECT category, COUNT(DISTINCT customer_id) AS UniqueCustomers
129 FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
130 GROUP BY category
131 ORDER BY UniqueCustomers DESC;

```

100 % No issues found

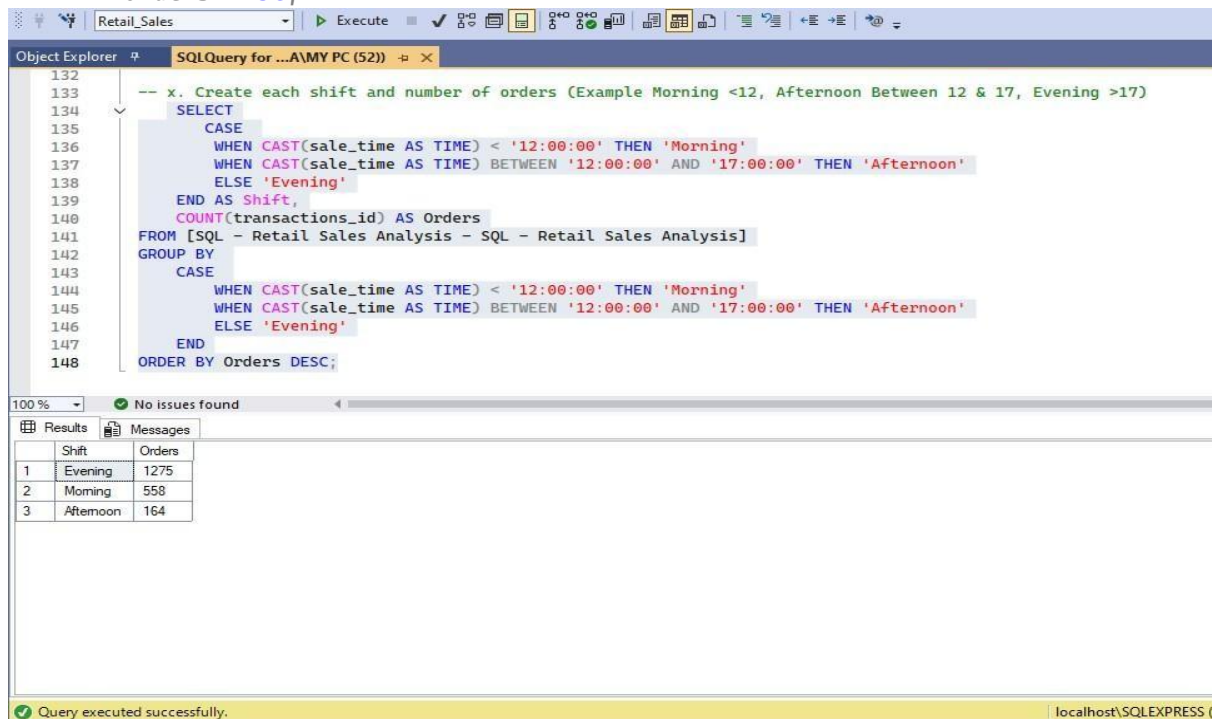
Results Messages

	category	UniqueCustomers
1	Clothing	149
2	Electronics	144
3	Beauty	141

Query executed successfully.

X. Create each shift and number of orders (Example Morning <12, Afternoon Between 12 & 17, Evening >17)

```
SELECT
CASE
    WHEN CAST(sale_time AS TIME) < '12:00:00' THEN 'Morning'
    WHEN CAST(sale_time AS TIME) BETWEEN '12:00:00' AND '17:00:00' THEN 'Afternoon'
    ELSE 'Evening'
END AS Shift,
COUNT(transactions_id) AS Orders
FROM [SQL - Retail Sales Analysis - SQL - Retail Sales Analysis]
GROUP BY
CASE
    WHEN CAST(sale_time AS TIME) < '12:00:00' THEN 'Morning'
    WHEN CAST(sale_time AS TIME) BETWEEN '12:00:00' AND '17:00:00' THEN 'Afternoon'
    ELSE 'Evening'
END
ORDER BY Orders DESC;
```



The screenshot shows the SQL Server Enterprise Manager interface. The top pane displays the SQL query being executed. The bottom pane shows the results of the query, which is a table with two columns: Shift and Orders. The table contains three rows of data, ordered by the number of orders in descending order.

Shift	Orders
Evening	1275
Morning	558
Afternoon	164

KEY FINDINGS:

- The data-quality check also found missing values before the analysis. There were 10 missing age records and three missing records each for quantity, price per unit, COGS and total sales. Age was handled by adding an age_flag to distinguish known ages from missing ages, while records missing key sales fields were removed.
- The analysis showed that Electronics recorded the highest total sales at 313,810, followed closely by Clothing at 311,070. Beauty recorded the lowest total sales at 286,450. Electronics and Clothing were the strongest categories by total sales, their sales performance makes them important categories to maintain and monitor.
- Beauty recorded the lowest total sales at 286,450 and had 624 transactions, compared with 701 for Clothing and 698 for Electronics. The Beauty category generated a higher average value per transaction but had fewer

transactions overall. Increasing the number of Beauty transactions could therefore have a meaningful effect on its total sales.

- December was the highest-selling month in both years. Total monthly sales were 72,880 in December 2022 and 69,145 in December 2023. Sales were particularly strong toward the end of each year.
- The five highest-spending customers were identified, with Customer_Id 3 recording the highest total spending at 38,440.
- Clothing had the highest number of unique customers at 145, followed by Electronics with 144 and Beauty with 141. The difference was small, indicating that the three categories attracted a similar number of unique customers even though their total sales differed.
- The time-of-day analysis showed a clear concentration of orders in the evening. There were 1,275 evening orders, compared with 555 in the morning and 164 in the afternoon. This means the evening was the busiest period.

RECOMMENDATIONS:

- **Improve data quality checks before analysis.**

The missing values found in the dataset show the importance of checking data quality before carrying out analysis.

- **Maintain the strong sales performance of Electronics and Clothing.**

Ensure adequate stock of products in these categories and give them prominent placement in sales channels during periods of high customer activity. Promotional offers can be used selectively on products that need to move faster rather than applying discounts across the entire category.

- **Increase the number of Beauty transactions.**

Offer promotional activities like Bundle offers, Buy-more incentives to increase the number of Beauty purchases rather than reducing prices across the category

- **December increase in sales.**

For December sales period, there should be sufficient inventory of products and increase promotional activities around products that are appropriate for seasonal purchasing like Holiday-season product bundles, Gift-oriented product promotions

- **Retain and engage high-spending customers.**

Retain high-spending customers through practical customer-retention measures such as: Loyalty rewards based on actual purchase activity, targeted offers on products or categories they regularly purchase

- **Match operations to evening demand.**

Ensure that products are available and customer-service capacity is sufficient during the evening since it is the busiest period.

Click here for the SQL file

https://drive.google.com/file/d/1PslvOQHUlyynX3v8L3_kEiMKyZfhFhJR/view?usp=sharing

